

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0770

Henry County 230 kV
September 22, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0770. The Interconnection Customer proposes a 155.1 MW solar photovoltaic facility interconnecting at the Henry County 230 kV substation in the Indiana area. The request was received on April 02, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 298 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,333,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Omni-Hackberry 500 kV/230 kV TX	Norwich to Omni-Hackberry 500.0 kV Line	101.5	942	7.9
Lighthouse Hill to Aes Beaver Vly Partners Beaver Vly 345.0 kV	Hilltop to Piney Crk Project 500.0 kV	101.0	1,193	7.2
Stewart Mtn Dam to Lighthouse Hill 345.0 kV	Piney Creek Project to Hilltop 500.0 kV line	100.0	1,059	7.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Omni-Hackberry Substation 500/230 kV Transformer	Replace transformer with higher-rated unit	\$48,901,000	2% (\$886,000)
Lighthouse Hill to Aes Beaver Vly Partners Beaver Vly 345.0 kV	Reconductor line	\$5,600,000	2% (\$119,000)
Stewart Mtn Dam to Lighthouse Hill 345.0 kV	Reconductor line	\$15,385,000	2% (\$328,000)
	Total allocated to project		\$1,333,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.